What Drives a Country's Innovation?

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1 Introduction

Patent filings are often treated as concrete, measurable indicators of innovation, yet their distribution across countries and years is strikingly uneven. Some economies are able to translate population size, urban structure, and technological diffusion into sustained inventive activity, whereas others fail to do so, despite sharing similar indicators.

In this research, we're asking: how well can we predict the intensity of patenting in a given year, considering a country's demographic, socio-economic, and technological indicating factors, such as urbanization density? By framing patenting as a quantitative outcome shaped by population structure, connectivity, and innovation capacity, this study aims to clarify which structural features are most tightly associated with where and when new ideas are most likely to be formally codified.

This study remains purely statistical, and the results are interpreted as correlations rather than causal effects.

2 Dataset

2.1 Source and variables

For our analysis, we relied on datasets from the *United Nations Data Retrieval System*. To ensure cross-country comparability and maximize coverage, we focused on 2022, the most recent year with relatively few missing observations across sources. The variable Urban_perc was only available up to 2018. For **RandD** and **Researchers**, remaining gaps were addressed by carrying forward the most recent observation available for each country (up to 2021/2020 for Researchers and up to 2018 for RandD; see the R script for details). This temporal misalignment is unlikely to materially affect the results, as these indicators typically change gradually over short time windows. We then merged all the desired variables into a first dataset, which contains the following columns:

Patents: number of patent resident filings, per million population[1]

Country: names of each country in study

GDP: GDP per capita (US dollars, $)[4]

Density: density of population, per km^2[2]

Urban_perc: percentage of population living in urban areas[3]

Internet_perc: percentage of individuals using internet[6]

RandD: gross domestic expenditure on Research and Development, as a percentage of GDP[5]

Researchers: number of researchers, per million inhabitants[5]

Over60: percentage of population aged 60+ years[2]

2.2 Data exploration and visualization

As part of the preliminary data exploration, we examined the distribution of **Patents**, which was found to be highly right-skewed and characterized by extreme values. To address skewness and stabilize variance, we computed a Box-Cox transformation (after shifting the response variable to ensure positivity), and obtained an estimated lambda value of $\lambda \approx 0.05$, i.e., close to a log-type transform. For simplicity, we therefore used **log(Patents+1)** as our response variable in order to address the heavy tailed nature of the distribution, at the expense of direct interpretability on the original scale.

As shown by the histogram after the transformation, the distribution remains strongly concentrated at low levels of patent activity. This pattern is consistent with the highly uneven distribution of patenting activity across countries: a small group of economies accounts for a disproportionate share of global patent filings, while many countries show persistent low patenting intensity.

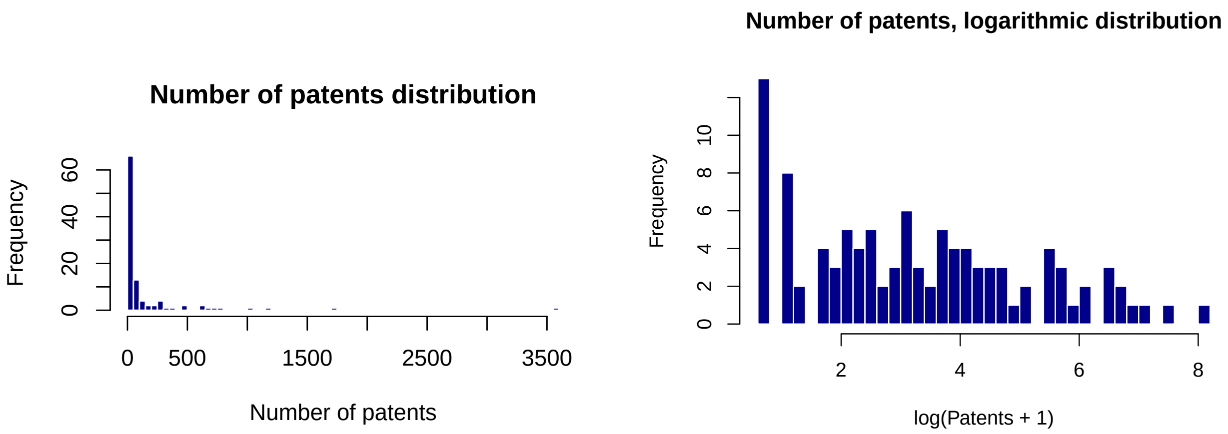
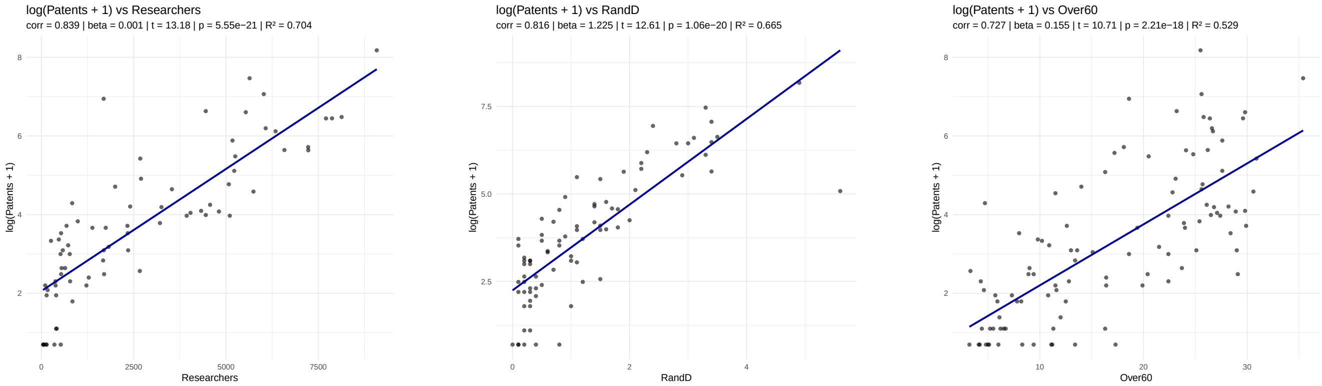


Figure 1: Distribution of patent intensity before and after log transformation.



(a) Researchers (b) R&D (c) Over 60

Figure 2: Log patent intensity vs. selected covariates.

Next, we plotted our (log-transformed) dependent variable against every other independent one to spot interesting linear relations and possible outliers: we identified some heavily correlated factors, like **RandD** or **Over60**, and some poorly correlated ones, like Density.

Log-transformation of population density. The **Density** variable shows substantial cross-country dispersion and, similarly to Patents, is characterized by a long right tail: a small number of countries are extremely dense, while the majority of states are comparatively sparse. Using density on the original scale would therefore compress most observations into a narrow range, and give disproportionate influence to the highest density cases. In order to address this issue, we used log(Density + 1) to reduce skewness and improve numerical stability.

We then proceeded with a simple linear regression with each factor against the Patents variable, and for each observation, we computed externally studentized residuals, which measure the deviation of an observation from the fitted relationship. By an initial overview of the dataset, we detected a few highly innovative economies (namely, the Republic of Korea, China, Israel, and Japan) as outlying observations. However, we did not treat these countries as outliers: their values are substantively meaningful rather than misleading, reflecting the high concentration of patenting activity across a small set of countries. As a result, no countries were excluded from our final dataset.

3 Model selection

3.1 Objective and candidate estimators

Our objective was to maximize out-of-sample predictive accuracy for patenting intensity, measured by the Root Mean Squared Error (**RMSE**) on **log(Patents+1)**, to assess how well the models perform when facing unseen data. We compared an Ordinary Least Squares (**OLS**) regression with three regularized linear estimators implemented in the `glmnet` library: **LASSO** (ℓ_1 penalty), **Ridge regression** (ℓ_2 penalty), and **Elastic Net** (which combines ℓ_1 and ℓ_2). These methods penalize the size of estimated parameters, and are therefore well suited in settings with correlated predictors. In our data, regressions of each predictor on the remaining data show us a strong dependence

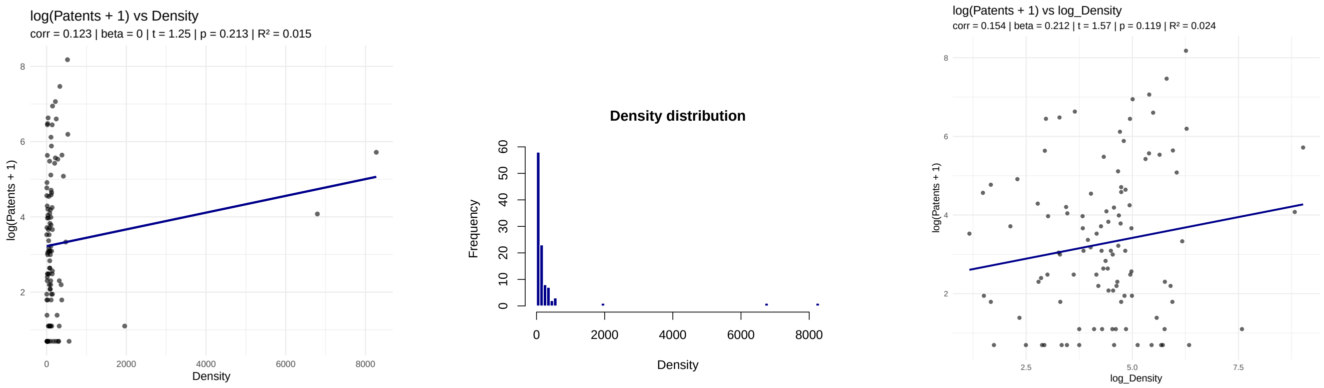


Figure 3: Adjustment of Density

among key covariates, as shown by the R_j^2 statistic (e.g., $R_j^2 = 0.882$ for **Researchers**, $R_j^2 = 0.779$ for **RandD**, and $R_j^2 = 0.667$ for **GDP**), suggesting that shrinkage may improve estimation stability and generalization.

3.2 Training protocol and hyperparameter tuning

All penalized models were tuned using 10-fold cross-validation within the training sample. For LASSO and Ridge, we selected the penalty parameter $\lambda_{\min}$, defined as the value minimizing the cross-validated error. For Elastic Net, we additionally tuned the mixing parameter $\alpha \in [0, 1]$ over a grid with step size 0.1, and selected the value of α that yielded the lowest cross-validated error; conditional on the chosen α, we set the penalty parameter to $\lambda_{\min}$ for predictions. This protocol ensures that model complexity is selected in a data-driven way and consistently across each estimator.

3.3 Repeated-split evaluation

Given the limited sample size (104 countries), model rankings based on a single train–test split may be noisy. We therefore evaluated predictive performance using repeated random splitting: observations are shuffled and partitioned into an 80% training set and a 20% test set, and the entire tuning-and-evaluation procedure is repeated 200 times. This repeated holdout design corresponds to a Monte Carlo evaluation (often referred to as *Monte Carlo cross-validation*[8]), as it approximates the out-of-sample error distribution by averaging performance over many random resamples. For each repetition, we computed the test-set RMSE for OLS, Ridge, LASSO, and Elastic Net.

3.4 Results and model choice

Across the 200 repetitions, Ridge regression achieved the lowest average RMSE (mean = 0.850, sd = 0.182), outperforming LASSO (mean = 0.869, sd = 0.172), Elastic Net (mean = 0.870, sd = 0.181), and OLS (mean = 0.881, sd = 0.178). In terms of split-wise dominance, Ridge is the best-performing method in 93 out of 200 splits (46.5%), compared with 48 (24.0%) for LASSO, 44 (22.0%) for OLS, and 15 (7.5%) for Elastic Net. Moreover, Ridge showed a lower RMSE than OLS in 58% of splits, indicating a consistent (though moderate) improvement in generalization. Overall, in this low-dimensional but correlated-feature setting, Ridge offered the most stable predictive accuracy, as it slightly increased bias while substantially reducing variance. We therefore retained Ridge (with $\lambda = \lambda_{\min}$) as our preferred predictive model.

4 Ridge regression (final specification and coefficients)

After selecting the model, we applied it on the full cleaned sample using the `cv.glmnet` function with 10-fold cross-validation. The estimated coefficients at $\lambda_{\min}$ are:

$$\hat{\beta}_0 = 1.018,\ \hat{\beta}_{\text{GDP}} = 8.65 \times 10^{-6},\ \hat{\beta}_{\text{Urban}} = -0.00544,\ \hat{\beta}_{\text{Internet}} = 0.0155,$$
$$\hat{\beta}_{\text{R\&D}} = 0.778,\ \hat{\beta}_{\text{Researchers}} = 1.08 \times 10^{-4},\ \hat{\beta}_{\text{Over60}} = 0.0351,\ \hat{\beta}_{\log(\text{Density}+1)} = -0.0643.$$

We then graphically evaluated the model fit within the training sample by comparing predicted and observed values of **log(Patents+1)** for the fitted Ridge model.

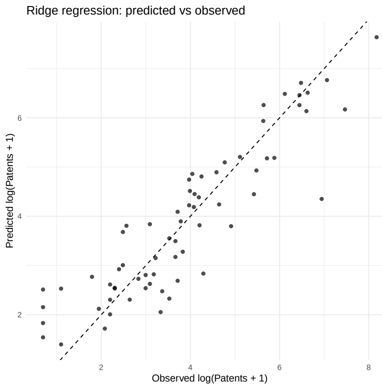


Figure 4: Ridge Prediction

5 Linear regression (OLS) and regression diagnostics

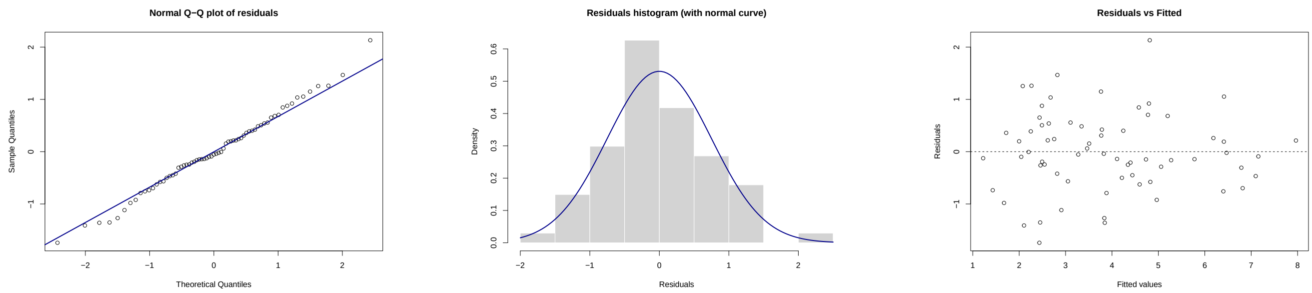
For a further quantitative comparison, we estimated the same linear specification using the Ordinary Least Squares regression via the `lm` function:

$$\log(\text{Patents}_i + 1) = \beta_0 + \beta_1\,\text{GDP}_i + \beta_2\,\text{Urban\_perc}_i + \beta_3\,\text{Internet\_perc}_i + \beta_4\,\text{RandD}_i + \\ + \beta_5\,\text{Researchers}_i + \beta_6\,\text{Over60}_i + \beta_7\,\log(\text{Density}_i + 1) + \varepsilon_i.$$

In the R script, this corresponds to:

```
m_diag <- lm(log_patents ~ GDP + Urban_perc + Internet_perc + RandD + Researchers
+ Over60 + log_Density, data = reduced).
```

We then evaluated whether the classical regression assumptions were broadly consistent with the data, using standard diagnostic plots and formal tests. Residual normality is strongly supported: the Shapiro-Wilk test yields $W = 0.9928$ with $p = 0.968$, and the Q–Q plot aligns closely with the reference line, indicating no meaningful departures from normality. To assess heteroskedasticity, we ran the studentized Breusch-Pagan test, obtaining $BP = 13.392$, with $df = 7$ and $p = 0.063$. This result is not significant at the 5% level but is close to conventional thresholds, suggesting at most mild evidence of heteroskedasticity. Consistently with these tests, we also inspected the residuals vs fitted plot for systematic patterns and produced a residual histogram, overlaid with a normal density curve. Overall, the diagnostics indicated that the OLS model is reasonably well-behaved in terms of residual normality, while variance constancy may be slightly imperfect, which further motivated the choice of a regularized estimator like Ridge for improved out-of-sample stability in the presence of correlated regressors.



(a) Normal Q-Q plot (b) Residuals histogram (c) Residuals vs fitted

Figure 5: OLS diagnostics for the baseline model: Q-Q plot, residual distribution, and residuals vs fitted values.

6 Does urbanization matter?

Given that urbanization is commonly included among the predictors of patenting activity, we assessed whether countries with different levels of urban population concentration show systematically different levels of patent intensity.

To this end, we divided countries into two groups based on the median share of population living in urban areas: a high-urbanization group and a low-urbanization group (see R script). We then compared average patenting intensity across the two groups, using a Welch two-sample t-test applied to **log(Patents + 1)**.

Let $Y_i = log(Patents_i + 1)$ and U_i be **Urban_perc** for country i; then our hypotheses will be:

$$H_0 : \mu_H \leq \mu_L \qquad and \qquad H_1 : \mu_H > \mu_L$$

where $\mu_H = E[Y_i|Urban_i > median(Urban)]$ and $\mu_L = E[Y_i|Urban_i <= median(Urban)]$: we are testing whether the mean patent activity differs between high and low urbanization countries.

The test yields a t-statistic of t=3.42 (with approximately 93.2 degrees of freedom) and p=0.00047. The one-sided 95% confidence interval for the difference in means is bounded below by 0.63, indicating a statistically significant gap in patenting intensity in favor of the high-urbanization group.

This result shows that urbanization is strongly associated with higher patenting activity in the raw data. While this comparison does not control for other covariates and therefore should not be interpreted causally, it provides a useful descriptive benchmark.

7 Conclusions and limitations

Our results indicate that a relatively small set of structural variables (**RandD, Researchers,** and **Internet_perc**) are strongly associated with patenting intensity. While population density shows some descriptive association with patent activity, its predictive contribution is comparatively weak with respect to other covariates. This suggests us that innovation outcomes are strongly linked to a country's research capacity and technological infrastructure, rather than to population concentration alone.

In this setting, regularization was revealed to be beneficial: across the train-test splits, Ridge regression consistently outperformed OLS, LASSO, and Elastic Net in terms of RMSE, offering more stable predictions in the presence of correlated regressors. This supports the use of shrinkage based estimators for our predictive model, considering both our moderate sample size and multicollinearity among our predictors.

Several limitations should be, however, acknowledged. First, our analysis relies on cross-sectional data for a single (mostly recent) year, which limits the possibility to capture dynamic innovation processes and lagged effects. In particular, variables such as **RandD** or **Researchers** may influence patenting with delays that cannot be modeled in a cross-sectional setting. Moreover, some indicators required temporal alignment adjustments due to data availability, which may have introduced additional measurement noise, even if such variables typically evolve slowly over time. Finally, patent filings are generally an imperfect proxy for innovation: their quality may vary across countries, just like the legal regimes under which they're released (which can affect their approval), and broadly not all innovative activity is patented. Furthermore, the observed associations reflect correlations in the data and may be driven by omitted institutional, cultural, or policy-related factors that are not captured in our study.

In order to take this analysis a step further, a more advanced model could be trained with data from multiple years, allowing for lagged predictors and country fixed effects. This would help separate persistent country characteristics from short-run changes. Another extension would be to explore alternative measures of innovative output, such as citations-weighted patents or sector-specific indicators, which could provide a more nuanced view of innovation performance.

8 Bibliography

[1] `https://data.un.org/_Docs/SYB/CSV/SYB67_264_202411_Patents.csv`

[2] `https://data.un.org/_Docs/SYB/CSV/SYB67_1_202411_Population,%20Surface%20Area%20and%20Density.csv`

[3] `https://data.un.org/_Docs/SYB/CSV/SYB61_253_Population%20Growth%20Rates%20in%20Urban%20areas%20and%20Capital%20cities.csv`

[4] `https://data.un.org/_Docs/SYB/CSV/SYB67_230_202411_GDP%20and%20GDP%20Per%20Capita.csv`

[5] `https://data.un.org/_Docs/SYB/CSV/SYB67_285_202411_Research%20and%20Development%20Expenditure%20and%20Staff.csv`

[6] `https://data.un.org/_Docs/SYB/CSV/SYB67_314_202411_Internet%20Usage.csv`

[7] Bijma F. et al. (2016). "*An Introduction to Mathematical Statistics*"

[8] `https://en.wikipedia.org/wiki/Cross-validation_(statistics)#Monte_Carlo_cross-validation`